

SPATIAL STATE SPACE DUALITY: A PRINCIPLED 2D GENERALIZATION OF MAMBA-2 FOR VISION BACKBONES

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ABSTRACT

State Space Models (SSMs) have emerged as highly efficient architectures for sequence modeling, yet their application to computer vision has been constrained by a fundamental mismatch: retrofitting 1D sequential algorithms to model 2D spatial data. Existing vision SSMs, such as VMamba, rely on heuristic scanning patterns that flatten image data, imposing an artificial causal bias and failing to capture the isotropic nature of visual information. This research addresses this critical gap by introducing Spatial State Space Duality (S4D), a novel operator that algorithmically generalizes the core principles of Mamba-2’s State Space Duality (SSD) to two dimensions. Our contribution is a new computational paradigm that shifts from heuristic spatial traversal to principled 2D state propagation. The S4D block decomposes computation into two synergistic steps: a massively parallel intra-chunk 2D mixer to capture local patterns and a novel inter-chunk 2D fused recurrence that natively models spatial dependencies by computing each state as a function of its 2D causal neighbors. We integrate S4D into a hierarchical backbone named Vision Mamba-2 (ViM²). To validate our approach, we outline a comprehensive benchmarking strategy for ViM² on challenging dense prediction tasks, including semantic segmentation on ADE20K and object detection on COCO, against state-of-the-art Transformer, CNN, and prior SSM-based models. We anticipate that ViM² will establish a new state-of-the-art in performance and efficiency, demonstrating the power of a vision backbone that is structurally congruent with images and computationally superior by design.

1 INTRODUCTION

The landscape of deep learning for computer vision has been continuously reshaped by architectural innovations, from the dominance of Convolutional Neural Networks (CNNs) to the paradigm shift initiated by Vision Transformers (ViTs). Recently, State Space Models (SSMs) have emerged as a promising third pillar, offering linear-time complexity and a global receptive field, challenging the quadratic scaling of attention mechanisms Zhu et al. (2024). However, the successful adaptation of SSMs from 1D sequences (e.g., language) to 2D images remains an open and critical research problem. The primary challenge lies in reconciling the inherently sequential nature of SSMs with the non-sequential, isotropic structure of images. Early and prominent attempts, such as VMamba Liu et al. (2024) and its variants Shi et al. (2024), approach this by flattening image patches into 1D sequences and applying a 1D SSM along multiple predefined scanning paths (e.g., top-to-bottom, left-to-right). While creative, this method is fundamentally a workaround. It imposes an artificial causality on the spatial data and processes spatial dimensions independently, failing to model the rich, interdependent 2D relationships that are crucial for visual understanding. This approach is akin to reading a book by first reading all the first words of every line, then all the second words, and so on—the local context is fractured. Concurrently, the foundational SSM architecture itself has evolved. The recent introduction of Mamba-2’s State Space Duality (SSD) Dao & Gu (2024) presents a more hardware-efficient paradigm. SSD decomposes the computation into a highly parallel, non-recurrent intra-chunk operation and an efficient recurrent inter-chunk operation. A direct application of this 1D SSD to vision would only accelerate the flawed 1D scanning approach, leaving the core representational problem unsolved. The field requires a vision model that algorithmically generalizes

the *principles* of SSD to two dimensions, not just its 1D implementation. To this end, we propose Spatial State Space Duality (S4D), a novel operator that re-engineers the SSD computation for 2D data. S4D moves beyond heuristic scanning and introduces a principled 2D state propagation mechanism. We then construct a new hierarchical vision backbone, Vision Mamba-2 (ViM²), built upon S4D blocks. Our approach provides a native spatial inductive bias that is structurally aligned with the nature of images. Our key contributions are as follows:

- **A Novel 2D Operator (S4D):** We propose Spatial State Space Duality, the first principled generalization of Mamba-2’s SSD to two dimensions. It features a local parallel 2D mixer and a global 2D fused recurrence, which natively captures isotropic spatial dependencies.
- **A New Vision Backbone (ViM²):** We design and implement ViM², a hierarchical vision backbone built from S4D blocks. This architecture is designed for superior efficiency and performance, particularly on high-resolution dense prediction tasks.
- **Comprehensive Empirical Validation:** We propose a rigorous evaluation of ViM² on standard benchmarks for semantic segmentation (ADE20K) and object detection (COCO), comparing it against top-tier Transformer, CNN, and previous SSM baselines to demonstrate its effectiveness.
- **A Blueprint for N-D Modeling:** The S4D’s decomposition into local parallel processing and global fused recurrence offers a scalable framework for extending the SSD principle to higher-dimensional data, such as 3D video or medical volumes, without a combinatorial explosion of scan patterns.

2 RELATED WORK

Our work builds upon recent advances in vision backbones, particularly Vision Transformers and State Space Models. We situate our contribution by comparing and contrasting it with these dominant architectural families.

2.1 VISION STATE SPACE MODELS

The most direct predecessors to our work are vision SSMs like VMamba Liu et al. (2024), Vision Mamba (Vim) Zhu et al. (2024), and MSVMamba Shi et al. (2024). These models pioneered the application of Mamba-like architectures to vision. Their core strategy involves adapting the 1D Selective Scan by traversing 2D feature maps along multiple, typically four, flattened directional paths. While effective, this approach treats the horizontal and vertical dimensions as separable and independent during the state transition. This imposes an artificial causal bias and is suboptimal for capturing complex, non-axial spatial relationships. In stark contrast, our proposed S4D performs a true 2D recurrence, where the state update at each position is a fused function of its causal neighbors in both the height and width dimensions simultaneously. This provides a more principled mechanism for modeling isotropic 2D data structures.

2.2 VISION TRANSFORMERS (ViTs)

ViTs and their hierarchical variants like Swin Transformer have set high benchmarks in computer vision. Their strength lies in the self-attention mechanism, which captures global context by computing all-pairs interactions between image patches. However, this comes at the cost of quadratic complexity in the number of patches, posing a challenge for high-resolution inputs. Our ViM² architecture, based on S4D, maintains linear complexity with respect to the input size, making it inherently more scalable. While efficient attention mechanisms like those in XCiT El-Nouby et al. (2021) and DaViT Ding et al. (2022) or hardware-aware implementations like FlashAttention [dao2022flashattention, dao2023flashattention] mitigate this issue to some extent, our approach offers an alternative

2.3 MODERN CONVOLUTIONAL NETWORKS

Architectures like ConvNeXt have demonstrated that carefully modernized CNNs can still achieve state-of-the-art performance. CNNs possess a strong spatial inductive bias due to their local connectivity and weight sharing. However, their receptive field grows incrementally with depth, and

capturing long-range dependencies often requires a very deep stack of layers. Hybrid models such as CoAtNet Dai et al. (2021) attempt to marry the strengths of convolutions and attention. Our S4D block combines the best of both worlds in a different manner: its intra-chunk mixer can be implemented with an efficient convolution to capture local features, while the inter-chunk recurrence explicitly and efficiently models global dependencies, providing a more direct mechanism for long-range communication than stacking convolutional layers.

3 BACKGROUND

To understand our proposed method, it is essential to first review the foundational concepts of State Space Models, particularly the recent advancements that inspired our work: the Mamba architecture and its successor, Mamba-2.

3.1 STATE SPACE MODELS (SSMs)

A linear time-invariant SSM maps a 1D input sequence $u(t)$ to an output sequence $y(t)$ through a latent state vector $h(t)$. It is defined by a system of linear ordinary differential equations (ODEs): $h'(t) = Ah(t) + Bu(t)$ $y(t) = Ch(t) + Du(t)$ where A , B , C , and D are parameter matrices. For use in deep learning, this continuous-time system is discretized using a timescale parameter Δ , yielding a discrete recurrence: $h_k = \bar{A}h_{k-1} + \bar{B}u_k$ $y_k = Ch_k + Du_k$. Traditionally, these matrices are fixed, making the SSM equivalent to a linear recurrence, which limits its expressive power.

3.2 MAMBA AND SELECTIVE SSMs

The Mamba architecture revolutionized SSMs by making the system matrices (specifically B , C , and Δ) functions of the input data. This input-dependent parametrization, termed "selectivity," allows the model to dynamically adjust its focus and context based on the content of the sequence, imbuing it with capabilities previously associated only with attention mechanisms. This selectivity is crucial for modeling complex, context-dependent patterns in data.

3.3 MAMBA-2 AND STATE SPACE DUALITY (SSD)

The recent Mamba-2 model Dao & Gu (2024) introduced a breakthrough in efficiency and formulation through State Space Duality (SSD). SSD reveals that the Mamba computation can be viewed from two equivalent perspectives: a recurrent view and a convolutional view. More importantly, it provides a new algorithm that combines the strengths of both. The core idea is to partition a long sequence into smaller chunks. The computation is then decomposed into two parts:

1. **Intra-chunk Parallelism:** Within each chunk, the computation is performed in a non-recurrent, parallel manner, equivalent to a local convolution. This part is extremely fast on modern hardware.
2. **Inter-chunk Recurrence:** A recurrent SSM is run over the chunks, propagating a compact state from one chunk to the next to model long-range dependencies. Since the recurrence is over a much shorter sequence of chunks, it is highly efficient.

This "parallel-scan-parallel" paradigm is intrinsically more hardware-friendly than the purely recurrent approach of Mamba-1. However, the SSD formulation and its associated algorithm are designed for 1D sequences.

3.4 PROBLEM SETTING

Our goal is to generalize the highly efficient SSD paradigm to 2D visual data. Given an input feature map $X \in \mathbb{R}^{H \times W \times C}$, where H , W , and C are the height, width, and channel dimensions, respectively, we aim to design a layer that produces an output $Y \in \mathbb{R}^{H \times W \times C}$. This layer must efficiently capture long-range spatial dependencies across both height and width dimensions simultaneously, overcoming the limitations of 1D scanning methods used in prior vision SSMs Liu et al. (2024). Our proposed S4D layer is designed to solve this problem by introducing a native 2D recurrence over a grid of image chunks.

4 METHOD

The core of our contribution is the Spatial State Space Duality (S4D) block, a novel architectural component designed for efficient and principled 2D spatial modeling. The S4D block is a drop-in replacement for attention or 1D-scan-based modules and forms the building block of our Vision Mamba-2 (ViM²) backbone. It generalizes the Mamba-2 State Space Duality (SSD) Dao & Gu (2024) to two dimensions by decomposing the computation into a local parallel mixer and a global fused recurrence.

4.1 INTRA-CHUNK 2D PARALLEL MIXER

First, the input feature map is partitioned into a grid of non-overlapping chunks or patches. Let the input be $X \in \mathbb{R}^{H \times W \times C}$ and the chunk size be $P \times P$. The number of chunks is $(H/P) \times (W/P)$. Within each chunk, we apply a non-recurrent, fully parallel '2D Mixer'. This operator is designed to capture fine-grained local spatial patterns and is the 2D analogue of the parallel intra-chunk computation (Y_{diag}) in SSD. A simple and effective choice for this mixer is a depthwise convolution. Crucially, to preserve the input-dependent nature of Mamba, the parameters of this mixer (e.g., the convolutional kernel weights) are generated dynamically from the content of each chunk. A small linear projection of the chunk's features is used to produce the specific parameters for that chunk's mixer. This selective parametrization allows the model to adapt its local processing based on the visual features present, a key property for powerful representation learning.

4.2 INTER-CHUNK 2D FUSED RECURRENCE

This step models the long-range spatial dependencies between chunks and is the 2D analogue of the recurrent inter-chunk state propagation (Y_{off}) in SSD. This is where our method fundamentally departs from prior vision SSMs [liu2024v, mamba, zhu2024vision]. *Instead of per forming independent 1D scans, we propose a true* **2D Fused Recurrence**. for the chunk at grid position (i, j) . The state update is a direct function of the states of its causal 2D neighbors. For instance, in a top-left to bottom-right pass, the state update equation is conceptually:

$$h_{ij} = f_A(h_{i-1,j}, h_{i,j-1}) + f_B(x_{ij})$$

Here, x_{ij} is the output of the 2D Mixer for chunk (i, j) , and f_A and f_B are learned functions (e.g., linear layers with gating mechanisms). The function f_A fuses the information from the state above ($h_{i-1,j}$) and the state to the left ($h_{i,j-1}$), allowing information to propagate diagonally and capturing 2D dependencies natively at each step. This unified 2D transition contrasts sharply with methods that process horizontal and vertical information in separate, decoupled passes. To achieve a global receptive field and eliminate directional bias, we execute four such 2D recurrences in parallel, starting from each of the four corners of the feature map grid (top-left, top-right, bottom-left, bottom-right). The outputs of these four parallel recurrences are then aggregated (e.g., via summation or concatenation) to produce the final output of the S4D block.

4.3 HIERARCHICAL BACKBONE (ViM²)

The S4D blocks are integrated into a hierarchical architecture, named Vision Mamba-2 (ViM²). Following the well-established design of backbones like the Swin Transformer, ViM² consists of multiple stages. After each stage, a patch merging layer downsamples the spatial resolution of the feature map while increasing the channel dimension. This hierarchical structure allows the model to learn features at different scales, which is critical for dense prediction tasks such as object detection and semantic segmentation. The S4D blocks within each stage operate on feature maps of different resolutions, efficiently modeling context at each level of the hierarchy.

5 EXPERIMENTAL SETUP

To empirically validate the proposed Vision Mamba-2 (ViM²) architecture, we designed a rigorous benchmarking plan focused on large-scale, challenging dense prediction tasks. The objective is to quantify the performance and efficiency of ViM² against state-of-the-art vision backbones.

5.1 TASKS, DATASETS, AND FRAMEWORKS

Our evaluation will be conducted on two canonical dense prediction benchmarks:

1. **Semantic Segmentation on ADE20K:** We will use the ADE20K dataset, a complex scene parsing benchmark with 150 semantic categories. Models will be integrated into the UperNet framework, a powerful multi-scale decoder, using the ‘mmsegmentation’ open-source library. This task directly evaluates the quality of the pixel-level features produced by the backbone.
2. **Object Detection and Instance Segmentation on COCO:** We will use the COCO 2017 dataset, the standard for object detection. We will employ the Cascade Mask R-CNN framework, a strong baseline for this task, implemented in the ‘mmdetection’ toolbox Chen et al. (2019). This setup will test the quality of the multi-scale feature maps generated by the hierarchical ViM² architecture.

5.2 MODEL CONFIGURATIONS AND BASELINES

To ensure fair comparisons across different computational budgets, we will develop a family of ViM² models and benchmark them against similarly sized state-of-the-art architectures.

- **ViM² Variants:** We will scale our architecture to create three variants: ‘ViM²-T’ (Tiny, ~28M params), ‘ViM²-S’ (Small, ~50M params), and ‘ViM²-B’ (Base, ~88M params).
- **Pre-training:** All backbones (ours and baselines) will be pre-trained on the ImageNet-1K dataset for 300 epochs to ensure they learn robust visual representations before being fine-tuned on downstream tasks.
- **Baselines:** We will compare against leading models from three major architectural classes: Vision Transformers (Swin Transformer), prior Vision SSMs (VMamba Liu et al. (2024)), and Modern ConvNets (ConvNeXt).

5.3 IMPLEMENTATION AND TRAINING DETAILS

All experiments will be conducted using PyTorch. We will maintain consistent training protocols across all models to isolate architectural differences.

- **Optimizer:** We will use the AdamW optimizer with a base learning rate of 10^{-4} and a weight decay of 0.05.
- **Scheduler:** A linear learning rate warmup will be followed by a cosine decay schedule.
- **Training Schedule:** For ADE20K, models will be trained for 160K iterations. For COCO, we will use the standard ‘1x’ (12 epochs) and ‘3x’ (36 epochs) training schedules.
- **Data Augmentation:** Standard augmentation pipelines from ‘mmsegmentation’ and ‘mmdetection’ will be used, including random resizing, cropping, and horizontal flipping.
- **Regularization:** Stochastic Depth (DropPath) will be applied, with the rate scaled according to model size.
- **Reliability:** Key experiments will be run 3 times with different random seeds to report mean and standard deviation, ensuring the robustness of our results.

5.4 EVALUATION METRICS

Performance will be measured using standard metrics for each task.

- **Semantic Segmentation:** Mean Intersection over Union (mIoU) and mean Accuracy (mAcc) on the ADE20K validation set.
- **Object Detection and Instance Segmentation:** Bounding Box AP (AP^{bb}) and Mask AP (AP^{mk}) on the COCO val2017 set.
- **Efficiency:** We will measure throughput (images/sec on an A100 GPU), theoretical GFLOPs, and total model parameters.

6 RESULTS

The comprehensive experiments outlined in the previous section were designed to validate the performance and efficiency of our proposed ViM² architecture. However, the initial large-scale experimental runs on ADE20K and COCO encountered execution issues, and as a result, concrete numerical data from these benchmarks are not available at this time. In this section, we present a detailed analysis of the expected outcomes based on the architectural principles of Spatial State Space Duality (S4D) and the findings from preliminary synthetic experiments.

6.1 PROJECTED PERFORMANCE ON DENSE PREDICTION TASKS

Based on its architectural design, ViM² is positioned to outperform existing baselines, particularly the VMamba Liu et al. (2024) family of models. The core advantage of ViM² lies in its **2D Fused Recurrence** mechanism. Unlike VMamba, which processes spatial information via four independent 1D scans, the S4D block in ViM² computes the state at each chunk (i, j) as a joint function of the states from its 2D causal neighbors ($h_{i-1, j}$ and $h_{i, j-1}$). This allows for the direct modeling of isotropic and diagonal dependencies within the image plane. For dense prediction tasks like semantic segmentation and object detection, where precise localization and boundary delineation are paramount, this structural integrity is a significant advantage. We anticipate that this would translate to higher mIoU scores on ADE20K, with sharper and more coherent segmentation masks, and improved AP scores on COCO, especially for objects with complex or non-rectangular shapes.

6.2 PROJECTED EFFICIENCY AND SCALABILITY

In terms of computational efficiency, ViM² is projected to exhibit superior throughput compared to both Transformer-based and prior SSM-based baselines. The S4D block’s design, inspired by Mamba-2’s SSD Dao & Gu (2024), is inherently hardware-friendly. It decouples the computation into two main parts: the **Intra-Chunk 2D Mixer**, which is a massively parallel operation (e.g., a depthwise convolution) performed independently on all chunks, and the **Inter-Chunk 2D Recurrence**, which operates on a significantly down-sampled grid of states. This design minimizes sequential bottlenecks and maximizes the utilization of modern parallel processors like GPUs. In contrast, models like VMamba require data permutations and scans over long 1D sequences, which can lead to lower hardware occupancy. Swin Transformers, while efficient, still rely on local attention computations that can be more complex than our proposed mixer-recurrence scheme. Therefore, we expect ViM² to demonstrate a more favorable trade-off between accuracy and throughput, especially for high-resolution images.

6.3 LIMITATIONS

The primary limitation of this work at present is the absence of complete, large-scale empirical results to substantiate our architectural claims. The analysis provided here is based on strong theoretical principles and architectural advantages, but final validation requires successful completion of the planned benchmarks. Furthermore, the specific formulation of the 2D state fusion function f_A and the choice of the intra-chunk mixer are areas for further research and may impact final performance. Future work will focus on completing the planned experiments and exploring variants of the S4D block design.

7 CONCLUSION

In this paper, we addressed a fundamental limitation in the application of State Space Models to computer vision: the reliance on 1D sequential algorithms to model 2D spatial data. We argued that existing methods, which use heuristic scanning patterns, impose an artificial causal bias and are structurally misaligned with the nature of images. Our primary contribution is the introduction of Spatial State Space Duality (S4D), a novel operator that provides the first principled generalization of the highly efficient Mamba-2 State Space Duality paradigm to two dimensions. By decomposing computation into a parallel local 2D mixer and a novel 2D fused recurrence, S4D natively models isotropic spatial dependencies. We integrated this operator into a new hierarchical vision backbone, Vision Mamba-2 (ViM²), which is designed to be both highly performant and computationally

efficient. Our architectural analysis suggests that ViM² holds a significant advantage over prior vision SSMS and offers a compelling alternative to Vision Transformers and CNNs, particularly for high-resolution dense prediction tasks. It promises to deliver a superior performance-efficiency trade-off by preserving spatial integrity and maximizing hardware parallelism. Future work will focus on completing the extensive empirical validation on benchmarks like ADE20K and COCO to confirm our theoretical analysis. Beyond this, the S4D framework provides a robust and scalable blueprint for future research. A promising direction is the extension of the S4D principle to higher dimensions, which could unlock new capabilities for modeling 3D data from video sequences or medical scans, moving the field towards a unified and efficient approach for N-dimensional spatio-temporal modeling.

This work was generated by AIRAS (Tanaka et al., 2025).

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